INF-2201
06 – Processes, Threads, Scheduling, Preemption

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2026

Topic: Processes and multitasking

- Work in progress
  - Reorganizing lectures to match projects (moving target)

Problem: computer can only run one program simultaneously

- Computers used to be very expensive
- First thought: Sometimes the CPU is just waiting for I/O (user input, disks, paper tapes, somebody to load the tapes in the reader, ...)
  - What if you could have some other program running simultaneously that could do things in the background?
- Second thought: if you can run multiple programs at the same time, then multiple users can share the same computer

Idea: multitasking

- Multitasking: let a computer run multiple tasks concurrently
  - If only one CPU:
    - switch between tasks often enough to give the illusion of running them at the same time
    - Switch to a different task when
      - one is blocked / waiting for something
      - when the current task has run for some time and we need to let other tasks make some progress

Observation

What does the program use and keep track of for a running program?

Basic information (for the current simple computer):

- Instruction pointer (IP/EIP)
- Stack pointer
- Registers

How they are used

- Variables are typically stored on heap (global variables or allocated variables) or stack/registers (local variables). NB: assuming a C-like language.
- Progress of the program is tracked using the IP
- Function calls:
  - Store state you need to keep (ex: registers) on the stack
  - Store parameters to function on stack or in registers (here: register A)
  - Store return pointer on the stack (the next instruction / address after the "call")
  - Jump/Call to the function
    - Function looks at parameters and does its things
    - Return value from function may be stored in a register (can also use stack, but slightly more tricky)
    - When returning back to caller: pop the stored instruction pointer into current IP (ret, iret, ...)
  - Control returns to the calling function

```
; Super simple program in z80-ish asm
…
0d00  push b        ; store registers
0d01  push c        ; store registers
0d02  mov a, 42     ; set parameter
0d04  push 0d0a     ; return addr/ip
0d07  call 0f00     ; call foo
0d0a  pop c         ; return here
0d0b  pop b
……...

; "function foo"
0f00 inc a          ; do something
…                   ; do something
0f40  mov a, 90     ; set return value
0f42  ret           ; return / pop ip
```

Tracing the instructions executed draws a "thread" of execution through the program

Observation

Basic information (for the current simple computer):

- Instruction pointer (IP/EIP)
- Stack pointer
- Registers

How they are used

- Sufficient to keep track of the current state of a running program in the current system we have!
- Store much of them when calling a function
- What if we can store everything we need before calling a "super function" that swiches to a different program?
  - All registers
  - SP
  - IP

```
; Super simple program in z80-ish asm
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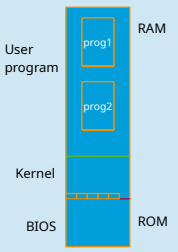
Tracing the instructions executed draws a "thread" of execution through the program

Running multiple programs

- Switch to another program (**yield**):
  - store/snapshot current state (all registers, including IP to "return here" and SP)
  - call function that selects which program gets to run next (let's call that the scheduler)
  - Restore registers (start with stack pointer)
  - "Return" to selected/scheduled program using "ret"
- NB: order of store and restore is important. One method:
  - Store IP of where to return to
  - Store registers
  - Store SP "somewhere"
- When restoring state
  - Restore SP (to pick the right stack)
  - Restore registers
  - "ret" to return to the indicated return position (instruction after the call to yield)



Processes

We currently have

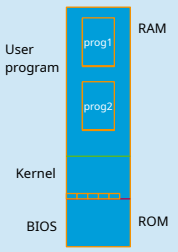
- Method for yielding execution to let somebody else take over: yield
- Method for selecting the next to run: scheduler

This is the first step to what we call **cooperative multitasking**

- Multiple programs that run concurrently, yielding execution time to each other

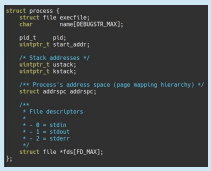
But how do we pick the next to run (**scheduler**), and where do we store the state of the programs?

- Introduce processes:
  - A process is a running program
    - The program binary
    - Data used by the program
    - Execution state (registers and stack)
    - Any other state that the operating system needs to keep track of the running process



Keeping track of processes

- Need information
  - Current execution state
    - When it is running: registers
    - When it is paused: kernels typically use a PCB (process control block) to save state necessary to restore the execution
  - Other state necessary for the kernel and processes
    - Identifier (ex: process id)
    - Permissions/right (we will look into this later)
- What is stored in a PCB? (Assume registers are stored on stack when yielding)
  - Process ID
  - Maybe memory location where the process is stored
  - Stack pointer (to let us restore/find stack and restore registers from there)
  - Some implementations might store a small kernel stack in the PCB to use for kernel functionality



Process struct from Munix P2 (2026)

Kernel needs to keep track of multiple processes

- If each process has a PCB that uniquely describes that process and state, use that PCB to also keep track of which processes that are currently waiting to run and which process is currently running
  - current_process → PCB of currently running process
  - ready_queue → queue of PCBs / processes waiting to run
- Process switching: when the **scheduler** is called:
  - Move the current_process PCB to the end of the ready queue
  - Move the first PCB from ready_queue to current_process
  - Restore process state of current_process
- This is called round robin scheduling
  - There are many options for more advanced scheduling algorithms, prioritizing things like interactivity, important tasks vs. less important tasks (priorities) etc.

How do we start a new process?

- We currently have mechanisms for yielding (storing state), selecting new state (schedule) and restore state.
- Instead of creating yet another mechanism, simply use the kernel's scheduler:
  - Create a restore state pointing to where you want the process to start
    "fake" a process that was interrupted before running its first instruction
  - Fill in the PCB
  - Submit the process to the kernel scheduler
  - Let the scheduler run, pick and activate a process

Problem: starting and stopping new processes

How do we create new processes from user level (from a process)?

- "Parent" calls fork()
  - Makes a copy of the current process.
  - New process (child) gets a new PCB. Starts executing at the same instruction as the parent
  - Identifying which copy is the parent or child: Return value of fork() is
    - Parent: PID (Process ID) of child
    - Child: 0

This assumes Unix/POSIX API (other operating systems have similar mechanisms)

```c
/* Source code from "main wait" on Ubuntu 22.04 */

#include <sys/wait.h>
#include <stdint.h>
#include <stdlib.h>
#include <unistd.h>
#include <stdio.h>

int main(int argc, char *argv[])
{
    pid_t cpid, w;
    int wstatus;

    cpid = fork();
    if (cpid == -1) {
        perror("fork");
        exit(EXIT_FAILURE);
    }

    if (cpid == 0) {            /* Code executed by child */
        printf("Child PID is %jd\n", (intmax_t) getpid());
        if (argc == 1)
            pause();                    /* Wait for signals */
        _exit(atoi(argv[1]));

    } else {                    /* Code executed by parent */
        do {
            w = waitpid(cpid, &wstatus, WUNTRACED | WCONTINUED);
            if (w == -1) {
                perror("waitpid");
                exit(EXIT_FAILURE);
            }

            if (WIFEXITED(wstatus)) {
                printf("exited, status=%d\n", WEXITSTATUS(wstatus));
            } else if (WIFSIGNALED(wstatus)) {
```

Problem: starting and stopping new processes

Waiting for a child process to terminate?

- Using one of the wait-calls (here: waitpid)

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                exit(EXIT_FAILURE);
            }

            if (WIFEXITED(wstatus)) {
                printf("exited, status=%d\n", WEXITSTATUS(wstatus));
            } else if (WIFSIGNALED(wstatus)) {
```

Problem: child should run a different program

Sometimes need to run a different program in the child than in the parent

- Solution: call exec() in child after returning from fork()
- Replaces the program code (and data) in the current program with new code and data
  - Typically starts running from main()
  - Can be called at any point in time (not just after fork)
-

Python exec example: replace Python process with /bin/ls

```
>>> import os
>>> os.execv("/bin/ls", ["/bin/ls", "/usr/local"])
bin  etc  games  include  lib  man  sbin  share  src
```

NOP

Problem: multiple activities in a single program

- Some system calls are blocking
  - "wait for user input"
  - "wait for file to be updated by another program"
- Each of these block would block the entire process
  - Ex: if program is waiting for file updates, it will not make any progress on user input
- Could support multiple activities by forking / using multiple processes
  - One process to handle each activity
- Several challenges:
  - Memory overhead (complete copy of program)
  - Communication overhead (processes need to coordinate, keep copies of shared state consistent, ...)
  - Context switch overhead (switch between processes)
  - Fork/join overhead

Solution: Threads

- Revisiting: thread of execution
- Like leaving a string to navigate a labyrinth (think Ariadne)
- What if we could have more than one (explorer)?

Image: Microsoft Copilot

```
; Super simple program in z80-ish asm
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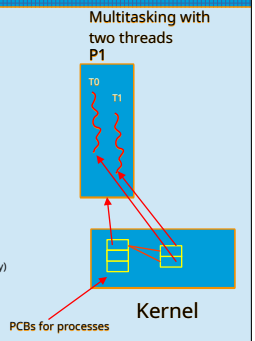
```

Tracing the instructions executed draws a "thread" of execution through the program

Solution: threads

Like processes, threads are used to keep track of execution. They differ in some ways:

- Parent and child thread share memory, open files etc.
- Use a different API (typical on Unix: pthreads), but similar concepts to process creation
- A thread is typically viewed as executing "inside" a process
  - One view: a process is a container that always has at least one default thread to track the execution. Can then add more threads.
  - Threads have individual Thread Control Blocks (TCB), but may share Process Control Blocks (PCB).
  - Less overhead for context switching
    - Some of it related to topics we haven't covered yet
  - Potentially less overhead for keeping shared state consistent (it's shared memory)
    - Some issues introduced though…



How do we run threads

We need

- some way of keeping track of each threads execution|
- Some mehanism for switching between threads
  - When do we switch?
- Mechanisms for starting, stopping and merging threads
  - One thread should be able to create more threads
  - The creator thread (parent) should be able to wait for the threads it created (child threads)
  - A thread should be able to finish / exit without killing other threads
    - What do we do when the last thread in a process exits?

Threads

Example: phtreads API

- Create a thread: pthread_create
  - Specify function to call and parameters to function
    - Where do you want the thread to start executing and what data should it get
  - Returns thread ID
- Wait for a thread: pthread_join
  - Specify thread ID
- What will this program print?

```c
#include <pthread.h>
#include <stdio.h>

const int N = 10;

typedef struct {
    int th_no;
} thread_parm;

void* thread_proc(void *vargs)
{
    thread_parm *parm = (thread_parm*) vargs;
    printf("This is thread with param %d\n", parm->th_no);
    return NULL;
}

int main(int argc, char* argv[])
{
    pthread_t threads[N];
    thread_parm parms[N];

    // Create N threads.
    for (int i = 0; i < N; i++) {
        parms[i].th_no = i;
        pthread_create(&threads[i], NULL, thread_proc, &parms[i]);
    }
    printf("All spawned, now waiting for them to complete\n");
    for (int i = 0; i < N; i++) {
        pthread_join(threads[i], NULL);
    }

    return 0;
}
```

Threads

Example: phtreads API

```
>cc -Wall thread1.c
>./a.out
This is thread with param 0
This is thread with param 3
This is thread with param 4
This is thread with param 1
This is thread with param 2
This is thread with param 5
This is thread with param 6
This is thread with param 7
This is thread with param 8
All spawned, now waiting for them to complete
This is thread with param 9
```

The system is not required to run the threads in the order you created them!

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    // Create N threads.
    for (int i = 0; i < N; i++) {
        parms[i].th_no = i;
        pthread_create(&threads[i], NULL, thread_proc, &parms[i]);
    }
    printf("All spawned, now waiting for them to complete\n");
    for (int i = 0; i < N; i++) {
        pthread_join(threads[i], NULL);
    }

    return 0;
}
```

How to implement threads

- User level threads – example with cooperative multitasking
  - Thread switch with function call (yield)
  - Should we even bother with this? Time?
  - Create, keep track of, schedule…
  - Issue: blocking system calls.
  - See more: coroutines, user space threads / user level threads
- Kernel threads – in Munix we will use **preemptive multitasking**:
  - Managed by kernel, not execute in kernel (confusion from old pre-2026 code)
  - Switching threads: already have mechanisms we can re-use: system calls and interrupts require storing and restoring current user level execution.
    - Key idea: if a thread is paused due to interrupt, what if we restore another thread instead?
    - Which thread to restore: decision made by scheduler.
    - When do we switch to give illusion of concurrent execution? Clock interrupt.
  - Scheduler:
    - Selects the next process or thread to run
    - Policy can be simple (round robin / next in queue), or more advanced (priorities, fair share, …, interactive vs. Batch, …)
  - Preempt: key idea is that the process or thread can be interrupted at any point in the execution, and the scheduler can pick another thread to run
    - Timer interrupt

Side note

- Linux has more flexibility when creating threads or processes when using the low-level **clone** system call
  - https://man7.org/linux/man-pages/man2/clone.2.html
- Lets caller control which resources should be shared or not shared between the parent and child.
  Whether it is a traditional process or thread depends on the set of flags specified when using clone.
-

Summary

- Multitasking
  - Cooperative vs. Preemptive multitasking
- Processes vs threads
- Scheduling
- Next week:
  - Concurrency, coordination, synchronization